

Exam rules

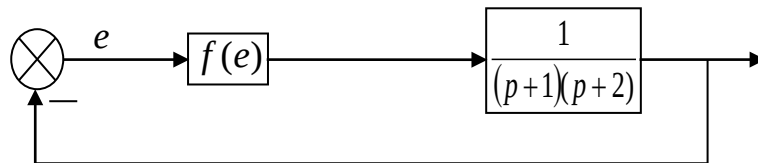
- 1) The exam takes 80 minutes. After finishing the exam, you must send me your solution within 15 minutes. After end of the exam, submitted work will not be considered.
- 2) *Retaking the exam takes 60 minutes (if you did not pass the exam the first time).*
- 3) The exam ticket number corresponds to your number in the group list.
- 4) The examination ticket consists of one theoretical task and one practical task. All tasks must be described in detail. On the first theoretical task, a theory must be written. For the second task, a detailed solution should be written.
- 5) Everyone sends me solutions in Ding Talk app. In Ding Talk, your name must be written in English letters (possible in a chat message).
- 6) Grades and scores are given according to the table below.

Result score (points)	Result grade	Result mark	Explanation	Comments
91 – 100	A	Pass	Excellent	Two tasks were completed correctly without errors.
84 – 90	B	Pass	Very good	One task is completed, and another task has minor errors.
75 – 83	C	Pass	Good	One task is completed, and another task has more than minor errors.
68 – 74	D	Pass	Satisfactory	One task is completed. The second task contains significant errors.
60 – 67	E	Pass	Satisfactory	Only one task is completed.
0 – 59	F	Fail	Unsatisfactory	None of the tasks have been completed.

Good luck!!!

Examination ticket № 1

- 1) Phase space method. Point transformation method.
- 2) Using Popov's frequency criterion, examine the system shown in the figure for stability.



The function $f(e)$ is presented as follows

$$f(e) = \begin{cases} 1, & e \geq 1, \\ e, & -1 < e < 1, \\ -1, & e \leq -1. \end{cases}$$

Examination ticket № 2

1) Lyapunov stability. The second Lyapunov theorem.

2) Using a circular criterion, investigate on absolute stability of the system in Lurie form. Define a sector $[k_1; k_2]$, in which the system is absolutely stable for the linear part

$$W(p) = \frac{1}{p^2 + 2p + 2}.$$

Examination ticket № 3

1) Absolute stability. Circular criterion.

2) A nonlinear system is given by

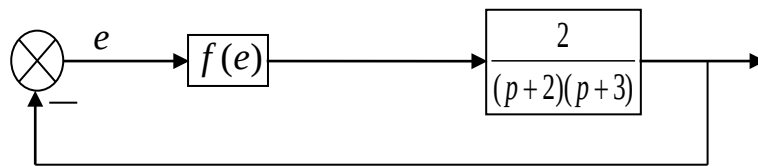
$$\begin{cases} \dot{x}(t) = y(t), \\ \dot{y}(t) = f(e(t)), \\ e(t) = -x(t). \end{cases} \quad f(e) = \begin{cases} 1, & e > 0,5, \\ 2e, & |e| < 0,5, \\ -1, & e < -0,5. \end{cases}$$

Draw the phase trajectories $(x(t), y(t) = \frac{dx(t)}{dt})$.

Examination ticket № 4

1) Absolute stability. Popov's criterion.

2) Using Popov's frequency criterion, investigate the system shown in the figure for absolute stability.



The function $f(e)$ is described as follows

$$f(e) = \begin{cases} 2, & e > 2, \\ 0, & -2 \leq e \leq 2, \\ -2, & e < -2. \end{cases}$$

Examination ticket № 5

1) Periodic solutions.

2) Using a circular criterion, investigate the absolute stability of the system in Lurie form. Define the sector $[k_1; k_2]$, in which the system is absolutely stable, where the linear part is represented by the transfer function

$$W(p) = \frac{1}{p^2 + 2p + 3}.$$

Examination ticket № 6

1) Self-oscillations (Periodic solution).

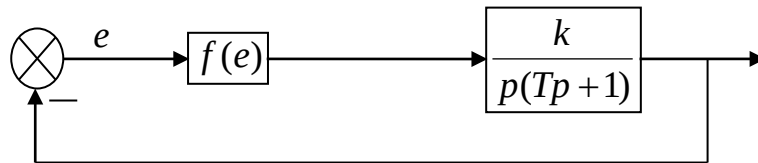
2) Draw a phase portrait for the system

$$\begin{cases} \dot{x}(t) = y(t), \\ \dot{y}(t) = f(e(t)), \\ e(t) = -x(t). \end{cases}$$

Here $f(e) = \begin{cases} 0, & -1 < e < 1, \\ 2\text{sign}(e), & |e| > 1. \end{cases}$

Examination ticket № 7

- 1) Systems with variable structure (sliding mode).
- 2) Using Popov's frequency criterion, investigate the system shown in the figure for absolute stability.



The function $f(e)$ is described as follows

$$f(e) = \begin{cases} 1, & e > 1, \\ 0, & -1 \leq e \leq 1, \\ -1, & e < -1. \end{cases}$$

Examination ticket № 8

- 1) Feedback control with providing inputs and outputs in given sets.
- 2) Using a circular criterion, investigate on absolute stability the system in Lurie form. Define the sector $[k_1; k_2]$, in which the system is absolutely stable, where the linear part is represented by the transfer function

$$W(p) = \frac{1}{p^2 + 3p + 4}.$$

Examination ticket № 9

- 1) Divergence method for stability study and stabilization of dynamical systems.
- 2) Draw a phase portrait for the system

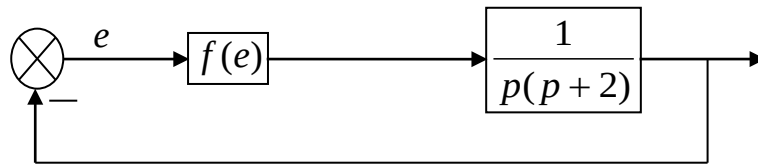
$$\begin{cases} \dot{x}(t) = y(t), \\ \dot{y}(t) = f(e(t)), \\ e(t) = -x(t). \end{cases}$$

Here $f(e)=3 \operatorname{sign}(e)$.

Examination ticket № 10

1) Control of Distributed Systems.

2) Using Popov's frequency criterion, investigate the system shown in the figure for absolute stability.



The function $f(e)$ is described as follows

$$f(e) = \begin{cases} 2, & e \geq 2, \\ e, & -2 < e < 2, \\ -2, & e \leq -2. \end{cases}$$

Examination ticket № 11

1) Phase space method. Point transformation method

2) Consider the system

$$\dot{x} = -x - x^3 - x^5.$$

Investigate the system for stability using the method of Lyapunov function.

Examination ticket № 12

1) Lyapunov stability. The second Lyapunov theorem.

2) Consider the system

$$\dot{x}_1 = -3x_1 + 7x_1^2$$

$$\dot{x}_2 = -x_2 + 2x_3^2$$

$$\dot{x}_3 = x_3 - 5x_1^2$$

Linearized system near zero equilibrium.

Examination ticket № 13

1) Absolute stability. Circular criterion.

2) Proof that the following system has no limit cycles

$$\dot{x}_1 = -x_1 + x_1^3 + x_1 x_2^2,$$

$$\dot{x}_2 = -x_2 + x_2^3 + x_1^2 x_2.$$

Examination ticket № 14

1) Absolute stability. Popov's criterion.

2) Consider a system described by a second-order equation

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = u$$

$$|u| \leq 1$$

The goal is to find the control law ensuring speed-optimal convergence to zero.

Examination ticket № 15

1) Periodic solutions.

2) Consider the system

$$\dot{x} = u + f(t)$$

where $f(t) = \sin(t)$. Investigate the control law

$$u = -a \operatorname{sgn}(x).$$

Find the value of a that guarantees Lyapunov stability.

Examination ticket № 16

1) Self-oscillations (Periodic solution).

2) Consider the system

$$\dot{x} = y, \quad \dot{y} = u$$

Investigate the control law

$$u = -\mu|y|^\alpha \operatorname{sgn} y - v|x|^{\frac{\alpha}{2-\alpha}} \operatorname{sgn} x$$

on Lyapunov stability of the closed-loop system.

Examination ticket № 17

1) Systems with variable structure (sliding mode).

2) Consider the system

$$\dot{x} = x^4 + u.$$

Investigate Lyapunov stability of the closed-loop system with the control law

$$u = -kx, \quad k > 0.$$

Examination ticket № 18

1) Feedback control with providing inputs and outputs in given sets.

2) Consider the system

$$\dot{x} = x^4 + u.$$

Investigate Lyapunov stability of the closed-loop system with the control law

$$u = -x^4 - x^3 - x.$$

Examination ticket № 19

1) Divergence method for stability study and stabilization of dynamical systems.

2) Consider the system

$$\dot{x}_1 = -x_1 + x_1 x_2^2 + u$$

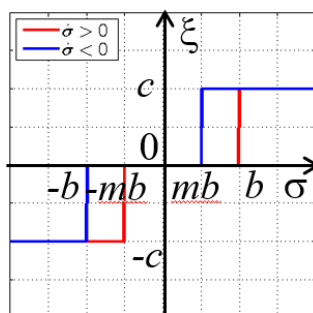
$$\dot{x}_2 = -3x_2^3 - x_2$$

Investigate Lyapunov stability of the closed-loop system with the control law

$$u = -kx.$$

Examination ticket № 20

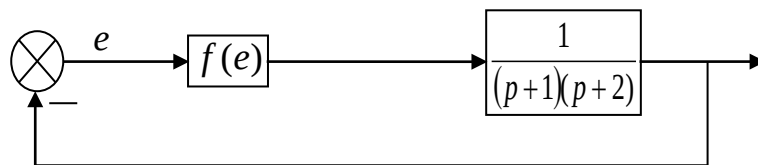
- 1) Control of Distributed Systems.
- 2) Consider a three-position relay



Using equivalent linearization method, find $g(a)$ and $b(a)$.

Examination ticket № 21

- 1) Phase space method. Point transformation method.
- 2) Using Popov's frequency criterion, examine the system shown in the figure for stability.



The function $f(e)$ is presented as follows

$$f(e) = \begin{cases} 1, & e \geq 1, \\ e, & -1 < e < 1, \\ -1, & e \leq -1. \end{cases}$$

Examination ticket № 22

1) Lyapunov stability. The second Lyapunov theorem.

2) Using a circular criterion, investigate on absolute stability of the system in Lurie form. Define a sector $[k_1; k_2]$, in which the system is absolutely stable for the linear part

$$W(p) = \frac{1}{p^2 + 2p + 2}.$$

Examination ticket № 23

1) Absolute stability. Circular criterion.

2) A nonlinear system is given by

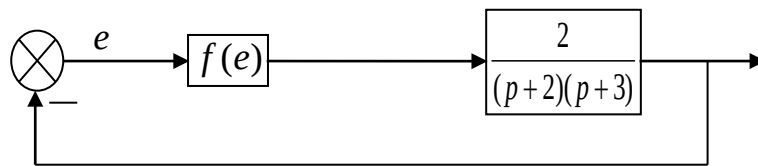
$$\begin{cases} \dot{x}(t) = y(t), \\ \dot{y}(t) = f(e(t)), \\ e(t) = -x(t). \end{cases} \quad f(e) = \begin{cases} 1, & e > 0,5, \\ 2e, & |e| < 0,5, \\ -1, & e < -0,5. \end{cases}$$

Draw the phase trajectories $\left(x(t), y(t) = \frac{dx(t)}{dt}\right)$.

Examination ticket № 24

1) Absolute stability. Popov's criterion.

2) Using Popov's frequency criterion, investigate the system shown in the figure for absolute stability.



The function $f(e)$ is described as follows

$$f(e) = \begin{cases} 2, & e > 2, \\ 0, & -2 \leq e \leq 2, \\ -2, & e < -2. \end{cases}$$

Examination ticket № 25

1) Periodic solutions.

2) Using a circular criterion, investigate the absolute stability of the system in Lurie form. Define the sector $[k_1; k_2]$, in which the system is absolutely stable, where the linear part is represented by the transfer function

$$W(p) = \frac{1}{p^2 + 2p + 3}.$$

Examination ticket № 26

1) Self-oscillations (Periodic solution).

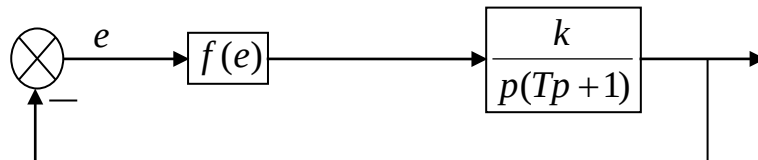
2) Draw a phase portrait for the system

$$\begin{cases} \dot{x}(t) = y(t), \\ \dot{y}(t) = f(e(t)), \\ e(t) = -x(t). \end{cases}$$

Here $f(e) = \begin{cases} 0, & -1 < e < 1, \\ 2\text{sign}(e), & |e| > 1. \end{cases}$

Examination ticket № 27

- 1) Systems with variable structure (sliding mode).
- 2) Using Popov's frequency criterion, investigate the system shown in the figure for absolute stability.



The function $f(e)$ is described as follows

$$f(e) = \begin{cases} 1, & e > 1, \\ 0, & -1 \leq e \leq 1, \\ -1, & e < -1. \end{cases}$$

Examination ticket № 28

- 1) Feedback control with providing inputs and outputs in given sets.
- 2) Using a circular criterion, investigate on absolute stability the system in Lurie form. Define the sector $[k_1; k_2]$, in which the system is absolutely stable, where the linear part is represented by the transfer function

$$W(p) = \frac{1}{p^2 + 3p + 4}.$$

Examination ticket № 29

- 1) Divergence method for stability study and stabilization of dynamical systems.
- 2) Draw a phase portrait for the system

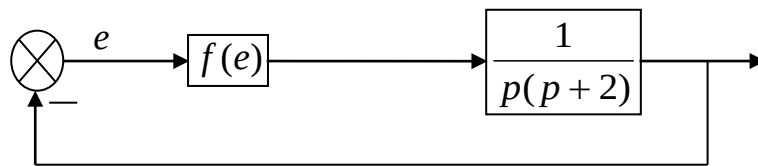
$$\begin{cases} \dot{x}(t) = y(t), \\ \dot{y}(t) = f(e(t)), \\ e(t) = -x(t). \end{cases}$$

Here $f(e)=3 \operatorname{sign}(e)$.

Examination ticket № 30

1) Control of Distributed Systems.

2) Using Popov's frequency criterion, investigate the system shown in the figure for absolute stability.



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$$f(e) = \begin{cases} 2, & e \geq 2, \\ e, & -2 < e < 2, \\ -2, & e \leq -2. \end{cases}$$

Examination ticket № 31

1) Phase space method. Point transformation method

2) Consider the system

$$\dot{x} = -x - x^3 - x^5.$$

Investigate the system for stability using the method of Lyapunov function.

Examination ticket № 32

1) Lyapunov stability. The second Lyapunov theorem.

2) Consider the system

$$\dot{x}_1 = -3x_1 + 7x_1^2$$

$$\dot{x}_2 = -x_2 + 2x_3^2$$

$$\dot{x}_3 = x_3 - 5x_1^2$$

Linearized system near zero equilibrium.

Examination ticket № 33

1) Absolute stability. Circular criterion.

2) Proof that the following system has no limit cycles

$$\dot{x}_1 = -x_1 + x_1^3 + x_1 x_2^2,$$

$$\dot{x}_2 = -x_2 + x_2^3 + x_1^2 x_2.$$

Examination ticket № 34

1) Absolute stability. Popov's criterion.

2) Consider a system described by a second-order equation

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = u$$

$$|u| \leq 1$$

The goal is to find the control law ensuring speed-optimal convergence to zero.

Examination ticket № 35

1) Periodic solutions.

2) Consider the system

$$\dot{x} = u + f(t)$$

where $f(t) = \sin(t)$. Investigate the control law

$$u = -a \operatorname{sgn}(x).$$

Find the value of a that guarantees Lyapunov stability.

Examination ticket № 36

1) Self-oscillations (Periodic solution).

2) Consider the system

$$\dot{x} = y, \quad \dot{y} = u$$

Investigate the control law

$$u = -\mu|y|^\alpha \operatorname{sgn} y - v|x|^{\frac{\alpha}{2-\alpha}} \operatorname{sgn} x$$

on Lyapunov stability of the closed-loop system.

Examination ticket № 37

1) Systems with variable structure (sliding mode).

2) Consider the system

$$\dot{x} = x^4 + u.$$

Investigate Lyapunov stability of the closed-loop system with the control law

$$u = -kx, \quad k > 0.$$

Examination ticket № 38

1) Feedback control with providing inputs and outputs in given sets.

2) Consider the system

$$\dot{x} = x^4 + u.$$

Investigate Lyapunov stability of the closed-loop system with the control law

$$u = -x^4 - x^3 - x.$$

Examination ticket № 39

1) Divergence method for stability study and stabilization of dynamical systems.

2) Consider the system

$$\dot{x}_1 = -x_1 + x_1 x_2^2 + u$$

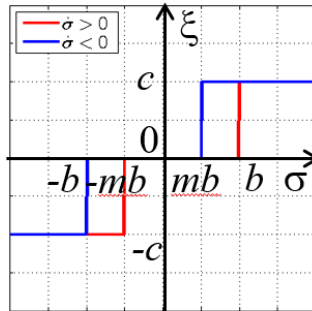
$$\dot{x}_2 = -3x_2^3 - x_2$$

Investigate Lyapunov stability of the closed-loop system with the control law

$$u = -kx.$$

Examination ticket № 40

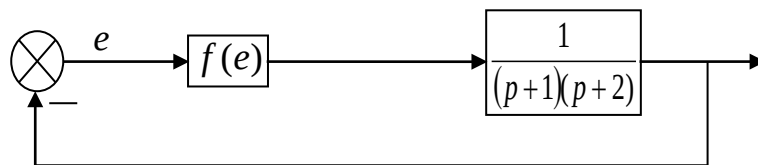
- 1) Control of Distributed Systems.
- 2) Consider a three-position relay



Using equivalent linearization method, find $g(a)$ and $b(a)$.

Examination ticket № 41

- 1) Phase space method. Point transformation method.
- 2) Using Popov's frequency criterion, examine the system shown in the figure for stability.



The function $f(e)$ is presented as follows

$$f(e) = \begin{cases} 1, & e \geq 1, \\ e, & -1 < e < 1, \\ -1, & e \leq -1. \end{cases}$$

Examination ticket № 42

1) Lyapunov stability. The second Lyapunov theorem.

2) Using a circular criterion, investigate on absolute stability of the system in Lurie form. Define a sector $[k_1; k_2]$, in which the system is absolutely stable for the linear part

$$W(p) = \frac{1}{p^2 + 2p + 2}.$$

Examination ticket № 43

1) Absolute stability. Circular criterion.

2) A nonlinear system is given by

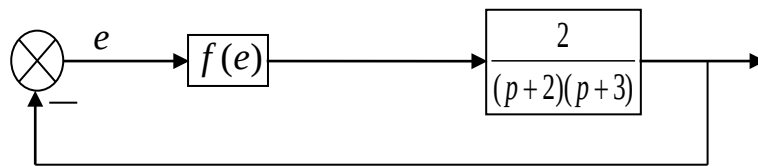
$$\begin{cases} \dot{x}(t) = y(t), \\ \dot{y}(t) = f(e(t)), \\ e(t) = -x(t). \end{cases} \quad f(e) = \begin{cases} 1, & e > 0,5, \\ 2e, & |e| < 0,5, \\ -1, & e < -0,5. \end{cases}$$

Draw the phase trajectories $\left(x(t), y(t) = \frac{dx(t)}{dt}\right)$.

Examination ticket № 44

1) Absolute stability. Popov's criterion.

2) Using Popov's frequency criterion, investigate the system shown in the figure for absolute stability.



The function $f(e)$ is described as follows

$$f(e) = \begin{cases} 2, & e > 2, \\ 0, & -2 \leq e \leq 2, \\ -2, & e < -2. \end{cases}$$

Examination ticket № 45

1) Periodic solutions.

2) Using a circular criterion, investigate the absolute stability of the system in Lurie form. Define the sector $[k_1; k_2]$, in which the system is absolutely stable, where the linear part is represented by the transfer function

$$W(p) = \frac{1}{p^2 + 2p + 3}.$$

Examination ticket № 46

1) Self-oscillations (Periodic solution).

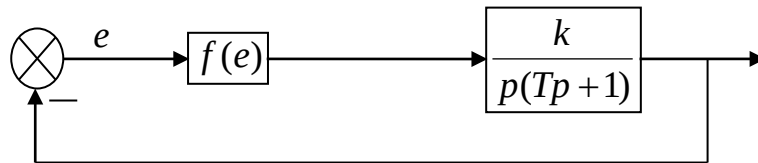
2) Draw a phase portrait for the system

$$\begin{cases} \dot{x}(t) = y(t), \\ \dot{y}(t) = f(e(t)), \\ e(t) = -x(t). \end{cases}$$

Here $f(e) = \begin{cases} 0, & -1 < e < 1, \\ 2\text{sign}(e), & |e| > 1. \end{cases}$

Examination ticket № 47

- 1) Systems with variable structure (sliding mode).
- 2) Using Popov's frequency criterion, investigate the system shown in the figure for absolute stability.



The function $f(e)$ is described as follows

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Examination ticket № 48

- 1) Feedback control with providing inputs and outputs in given sets.
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$$W(p) = \frac{1}{p^2 + 3p + 4}.$$

Examination ticket № 49

- 1) Divergence method for stability study and stabilization of dynamical systems.
- 2) Draw a phase portrait for the system

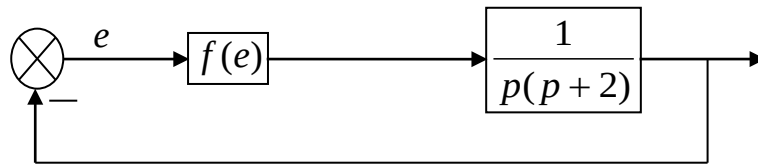
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Here $f(e)=3 \operatorname{sign}(e)$.

Examination ticket № 50

1) Control of Distributed Systems.

2) Using Popov's frequency criterion, investigate the system shown in the figure for absolute stability.



The function $f(e)$ is described as follows

$$f(e) = \begin{cases} 2, & e \geq 2, \\ e, & -2 < e < 2, \\ -2, & e \leq -2. \end{cases}$$

Examination ticket № 51

1) Phase space method. Point transformation method

2) Consider the system

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Investigate the system for stability using the method of Lyapunov function.

Examination ticket № 52

1) Lyapunov stability. The second Lyapunov theorem.

2) Consider the system

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Linearized system near zero equilibrium.

Examination ticket № 53

1) Absolute stability. Circular criterion.

2) Proof that the following system has no limit cycles

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Examination ticket № 54

1) Absolute stability. Popov's criterion.

2) Consider a system described by a second-order equation

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$$\dot{x}_2 = u$$

$$|u| \leq 1$$

The goal is to find the control law ensuring speed-optimal convergence to zero.

Examination ticket № 55

1) Periodic solutions.

2) Consider the system

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where $f(t) = \sin(t)$. Investigate the control law

$$u = -a \operatorname{sgn}(x).$$

Find the value of a that guarantees Lyapunov stability.

Examination ticket № 56

1) Self-oscillations (Periodic solution).

2) Consider the system

$$\dot{x} = y, \quad \dot{y} = u$$

Investigate the control law

$$u = -\mu|y|^\alpha \operatorname{sgn} y - v|x|^{\frac{\alpha}{2-\alpha}} \operatorname{sgn} x$$

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Examination ticket № 57

1) Systems with variable structure (sliding mode).

2) Consider the system

$$\dot{x} = x^4 + u.$$

Investigate Lyapunov stability of the closed-loop system with the control law

$$u = -kx, \quad k > 0.$$

Examination ticket № 58

1) Feedback control with providing inputs and outputs in given sets.

2) Consider the system

$$\dot{x} = x^4 + u.$$

Investigate Lyapunov stability of the closed-loop system with the control law

$$u = -x^4 - x^3 - x.$$

Examination ticket № 59

1) Divergence method for stability study and stabilization of dynamical systems.

2) Consider the system

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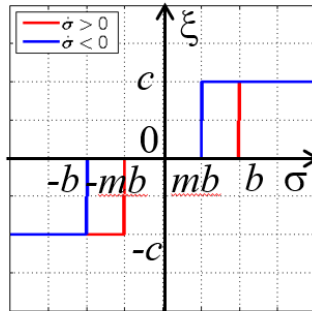
$$\dot{x}_2 = -3x_2^3 - x_2$$

Investigate Lyapunov stability of the closed-loop system with the control law

$$u = -kx.$$

Examination ticket № 60

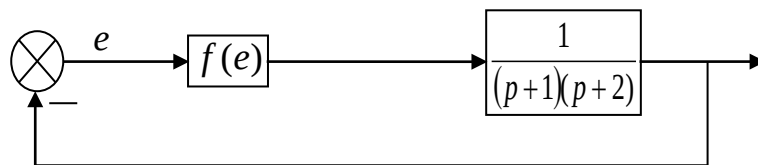
- 1) Control of Distributed Systems.
- 2) Consider a three-position relay



Using equivalent linearization method, find $g(a)$ and $b(a)$.

Examination ticket № 61

- 1) Phase space method. Point transformation method.
- 2) Using Popov's frequency criterion, examine the system shown in the figure for stability.



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1) Lyapunov stability. The second Lyapunov theorem.

2) Using a circular criterion, investigate on absolute stability of the system in Lurie form. Define a sector $[k_1; k_2]$, in which the system is absolutely stable for the linear part

$$W(p) = \frac{1}{p^2 + 2p + 2}.$$

Examination ticket № 63

1) Absolute stability. Circular criterion.

2) A nonlinear system is given by

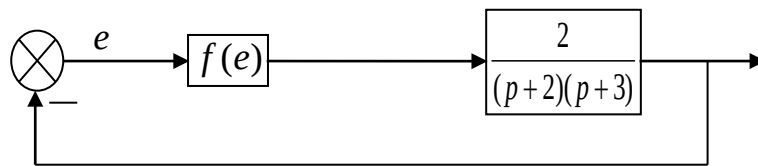
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Examination ticket № 64

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2) Using Popov's frequency criterion, investigate the system shown in the figure for absolute stability.



The function $f(e)$ is described as follows

$$f(e) = \begin{cases} 2, & e > 2, \\ 0, & -2 \leq e \leq 2, \\ -2, & e < -2. \end{cases}$$

Examination ticket № 65

1) Periodic solutions.

2) Using a circular criterion, investigate the absolute stability of the system in Lurie form. Define the sector $[k_1; k_2]$, in which the system is absolutely stable, where the linear part is represented by the transfer function

$$W(p) = \frac{1}{p^2 + 2p + 3}.$$

Examination ticket № 66

1) Self-oscillations (Periodic solution).

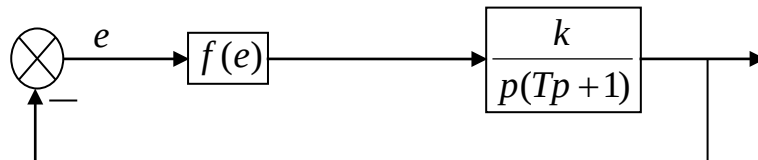
2) Draw a phase portrait for the system

$$\begin{cases} \dot{x}(t) = y(t), \\ \dot{y}(t) = f(e(t)), \\ e(t) = -x(t). \end{cases}$$

Here $f(e) = \begin{cases} 0, & -1 < e < 1, \\ 2\text{sign}(e), & |e| > 1. \end{cases}$

Examination ticket № 67

- 1) Systems with variable structure (sliding mode).
- 2) Using Popov's frequency criterion, investigate the system shown in the figure for absolute stability.



The function $f(e)$ is described as follows

$$f(e) = \begin{cases} 1, & e > 1, \\ 0, & -1 \leq e \leq 1, \\ -1, & e < -1. \end{cases}$$

Examination ticket № 68

- 1) Feedback control with providing inputs and outputs in given sets.
- 2) Using a circular criterion, investigate on absolute stability the system in Lurie form. Define the sector $[k_1; k_2]$, in which the system is absolutely stable, where the linear part is represented by the transfer function

$$W(p) = \frac{1}{p^2 + 3p + 4}.$$

Examination ticket № 69

- 1) Divergence method for stability study and stabilization of dynamical systems.
- 2) Draw a phase portrait for the system

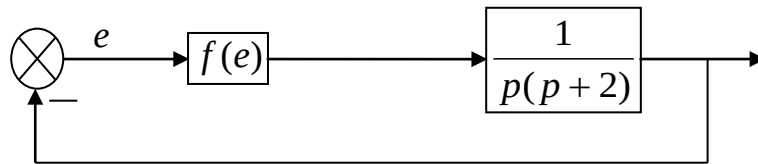
$$\begin{cases} \dot{x}(t) = y(t), \\ \dot{y}(t) = f(e(t)), \\ e(t) = -x(t). \end{cases}$$

Here $f(e)=3 \operatorname{sign}(e)$.

Examination ticket № 70

1) Control of Distributed Systems.

2) Using Popov's frequency criterion, investigate the system shown in the figure for absolute stability.



The function $f(e)$ is described as follows

$$f(e) = \begin{cases} 2, & e \geq 2, \\ e, & -2 < e < 2, \\ -2, & e \leq -2. \end{cases}$$

Examination ticket № 71

1) Phase space method. Point transformation method

2) Consider the system

$$\dot{x} = -x - x^3 - x^5.$$

Investigate the system for stability using the method of Lyapunov function.

Examination ticket № 72

1) Lyapunov stability. The second Lyapunov theorem.

2) Consider the system

$$\dot{x}_1 = -3x_1 + 7x_1^2$$

$$\dot{x}_2 = -x_2 + 2x_3^2$$

$$\dot{x}_3 = x_3 - 5x_1^2$$

Linearized system near zero equilibrium.

Examination ticket № 73

1) Absolute stability. Circular criterion.

2) Proof that the following system has no limit cycles

$$\dot{x}_1 = -x_1 + x_1^3 + x_1 x_2^2,$$

$$\dot{x}_2 = -x_2 + x_2^3 + x_1^2 x_2.$$

Examination ticket № 74

1) Absolute stability. Popov's criterion.

2) Consider a system described by a second-order equation

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = u$$

$$|u| \leq 1$$

The goal is to find the control law ensuring speed-optimal convergence to zero.

Examination ticket № 75

1) Periodic solutions.

2) Consider the system

$$\dot{x} = u + f(t)$$

where $f(t) = \sin(t)$. Investigate the control law

$$u = -a \operatorname{sgn}(x).$$

Find the value of a that guarantees Lyapunov stability.

Examination ticket № 76

1) Self-oscillations (Periodic solution).

2) Consider the system

$$\dot{x} = y, \quad \dot{y} = u$$

Investigate the control law

$$u = -\mu|y|^\alpha \operatorname{sgn} y - v|x|^{\frac{\alpha}{2-\alpha}} \operatorname{sgn} x$$

on Lyapunov stability of the closed-loop system.

Examination ticket № 77

1) Systems with variable structure (sliding mode).

2) Consider the system

$$\dot{x} = x^4 + u.$$

Investigate Lyapunov stability of the closed-loop system with the control law

$$u = -kx, \quad k > 0.$$

Examination ticket № 78

1) Feedback control with providing inputs and outputs in given sets.

2) Consider the system

$$\dot{x} = x^4 + u.$$

Investigate Lyapunov stability of the closed-loop system with the control law

$$u = -x^4 - x^3 - x.$$

Examination ticket № 79

1) Divergence method for stability study and stabilization of dynamical systems.

2) Consider the system

$$\dot{x}_1 = -x_1 + x_1 x_2^2 + u$$

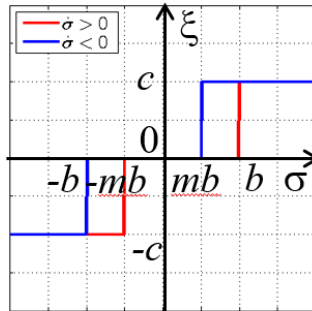
$$\dot{x}_2 = -3x_2^3 - x_2$$

Investigate Lyapunov stability of the closed-loop system with the control law

$$u = -kx.$$

Examination ticket № 80

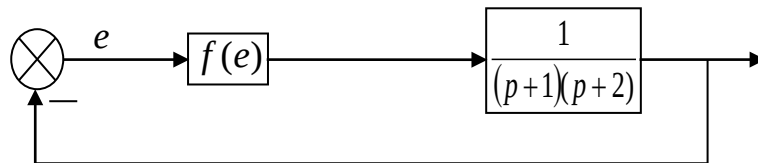
- 1) Control of Distributed Systems.
- 2) Consider a three-position relay



Using equivalent linearization method, find $g(a)$ and $b(a)$.

Examination ticket № 81

- 1) Phase space method. Point transformation method.
- 2) Using Popov's frequency criterion, examine the system shown in the figure for stability.



The function $f(e)$ is presented as follows

$$f(e) = \begin{cases} 1, & e \geq 1, \\ e, & -1 < e < 1, \\ -1, & e \leq -1. \end{cases}$$

Examination ticket № 82

1) Lyapunov stability. The second Lyapunov theorem.

2) Using a circular criterion, investigate on absolute stability of the system in Lurie form. Define a sector $[k_1; k_2]$, in which the system is absolutely stable for the linear part

$$W(p) = \frac{1}{p^2 + 2p + 2}.$$

Examination ticket № 83

1) Absolute stability. Circular criterion.

2) A nonlinear system is given by

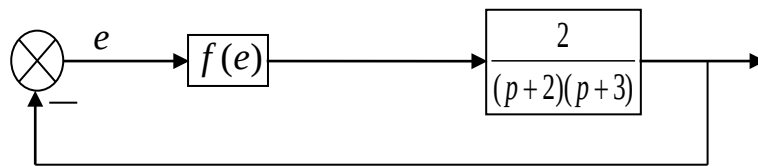
$$\begin{cases} \dot{x}(t) = y(t), \\ \dot{y}(t) = f(e(t)), \\ e(t) = -x(t). \end{cases}$$
$$f(e) = \begin{cases} 1, & e > 0,5, \\ 2e, & |e| < 0,5, \\ -1, & e < -0,5. \end{cases}$$

Draw the phase trajectories $(x(t), y(t) = \frac{dx(t)}{dt})$.

Examination ticket № 84

1) Absolute stability. Popov's criterion.

2) Using Popov's frequency criterion, investigate the system shown in the figure for absolute stability.



The function $f(e)$ is described as follows

$$f(e) = \begin{cases} 2, & e > 2, \\ 0, & -2 \leq e \leq 2, \\ -2, & e < -2. \end{cases}$$

Examination ticket № 85

1) Periodic solutions.

2) Using a circular criterion, investigate the absolute stability of the system in Lurie form. Define the sector $[k_1; k_2]$, in which the system is absolutely stable, where the linear part is represented by the transfer function

$$W(p) = \frac{1}{p^2 + 2p + 3}.$$

Examination ticket № 86

1) Self-oscillations (Periodic solution).

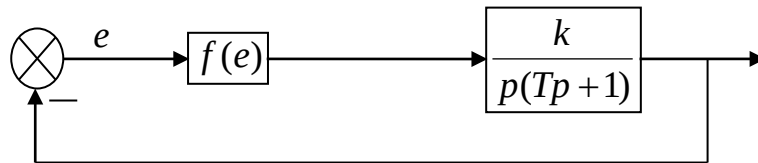
2) Draw a phase portrait for the system

$$\begin{cases} \dot{x}(t) = y(t), \\ \dot{y}(t) = f(e(t)), \\ e(t) = -x(t). \end{cases}$$

Here $f(e) = \begin{cases} 0, & -1 < e < 1, \\ 2\text{sign}(e), & |e| > 1. \end{cases}$

Examination ticket № 87

- 1) Systems with variable structure (sliding mode).
- 2) Using Popov's frequency criterion, investigate the system shown in the figure for absolute stability.



The function $f(e)$ is described as follows

$$f(e) = \begin{cases} 1, & e > 1, \\ 0, & -1 \leq e \leq 1, \\ -1, & e < -1. \end{cases}$$

Examination ticket № 88

- 1) Feedback control with providing inputs and outputs in given sets.
- 2) Using a circular criterion, investigate on absolute stability the system in Lurie form. Define the sector $[k_1; k_2]$, in which the system is absolutely stable, where the linear part is represented by the transfer function

$$W(p) = \frac{1}{p^2 + 3p + 4}.$$

Examination ticket № 89

- 1) Divergence method for stability study and stabilization of dynamical systems.
- 2) Draw a phase portrait for the system

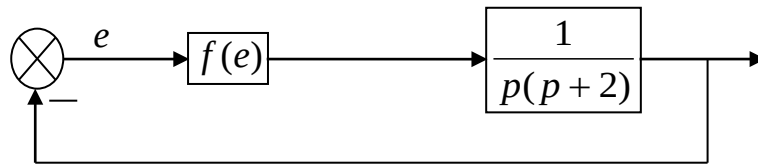
$$\begin{cases} \dot{x}(t) = y(t), \\ \dot{y}(t) = f(e(t)), \\ e(t) = -x(t). \end{cases}$$

Here $f(e)=3 \operatorname{sign}(e)$.

Examination ticket № 90

1) Control of Distributed Systems.

2) Using Popov's frequency criterion, investigate the system shown in the figure for absolute stability.



The function $f(e)$ is described as follows

$$f(e) = \begin{cases} 2, & e \geq 2, \\ e, & -2 < e < 2, \\ -2, & e \leq -2. \end{cases}$$

Examination ticket № 91

1) Phase space method. Point transformation method

2) Consider the system

$$\dot{x} = -x - x^3 - x^5.$$

Investigate the system for stability using the method of Lyapunov function.

Examination ticket № 92

1) Lyapunov stability. The second Lyapunov theorem.

2) Consider the system

$$\dot{x}_1 = -3x_1 + 7x_1^2$$

$$\dot{x}_2 = -x_2 + 2x_3^2$$

$$\dot{x}_3 = x_3 - 5x_1^2$$

Linearized system near zero equilibrium.

Examination ticket № 93

1) Absolute stability. Circular criterion.

2) Proof that the following system has no limit cycles

$$\dot{x}_1 = -x_1 + x_1^3 + x_1 x_2^2,$$

$$\dot{x}_2 = -x_2 + x_2^3 + x_1^2 x_2.$$

Examination ticket № 94

1) Absolute stability. Popov's criterion.

2) Consider a system described by a second-order equation

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = u$$

$$|u| \leq 1$$

The goal is to find the control law ensuring speed-optimal convergence to zero.

Examination ticket № 95

1) Periodic solutions.

2) Consider the system

$$\dot{x} = u + f(t)$$

where $f(t) = \sin(t)$. Investigate the control law

$$u = -a \operatorname{sgn}(x).$$

Find the value of a that guarantees Lyapunov stability.

Examination ticket № 96

1) Self-oscillations (Periodic solution).

2) Consider the system

$$\dot{x} = y, \quad \dot{y} = u$$

Investigate the control law

$$u = -\mu|y|^\alpha \operatorname{sgn} y - v|x|^{\frac{\alpha}{2-\alpha}} \operatorname{sgn} x$$

on Lyapunov stability of the closed-loop system.

Examination ticket № 97

1) Systems with variable structure (sliding mode).

2) Consider the system

$$\dot{x} = x^4 + u.$$

Investigate Lyapunov stability of the closed-loop system with the control law

$$u = -kx, \quad k > 0.$$

Examination ticket № 98

1) Feedback control with providing inputs and outputs in given sets.

2) Consider the system

$$\dot{x} = x^4 + u.$$

Investigate Lyapunov stability of the closed-loop system with the control law

$$u = -x^4 - x^3 - x.$$

Examination ticket № 99

1) Divergence method for stability study and stabilization of dynamical systems.

2) Consider the system

$$\dot{x}_1 = -x_1 + x_1 x_2^2 + u$$

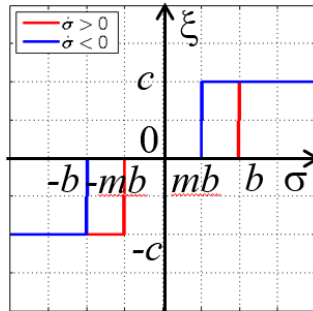
$$\dot{x}_2 = -3x_2^3 - x_2$$

Investigate Lyapunov stability of the closed-loop system with the control law

$$u = -kx.$$

Examination ticket № 100

- 1) Control of Distributed Systems.
- 2) Consider a three-position relay



Using equivalent linearization method, find $g(a)$ and $b(a)$.